



《CAD与工程制图》

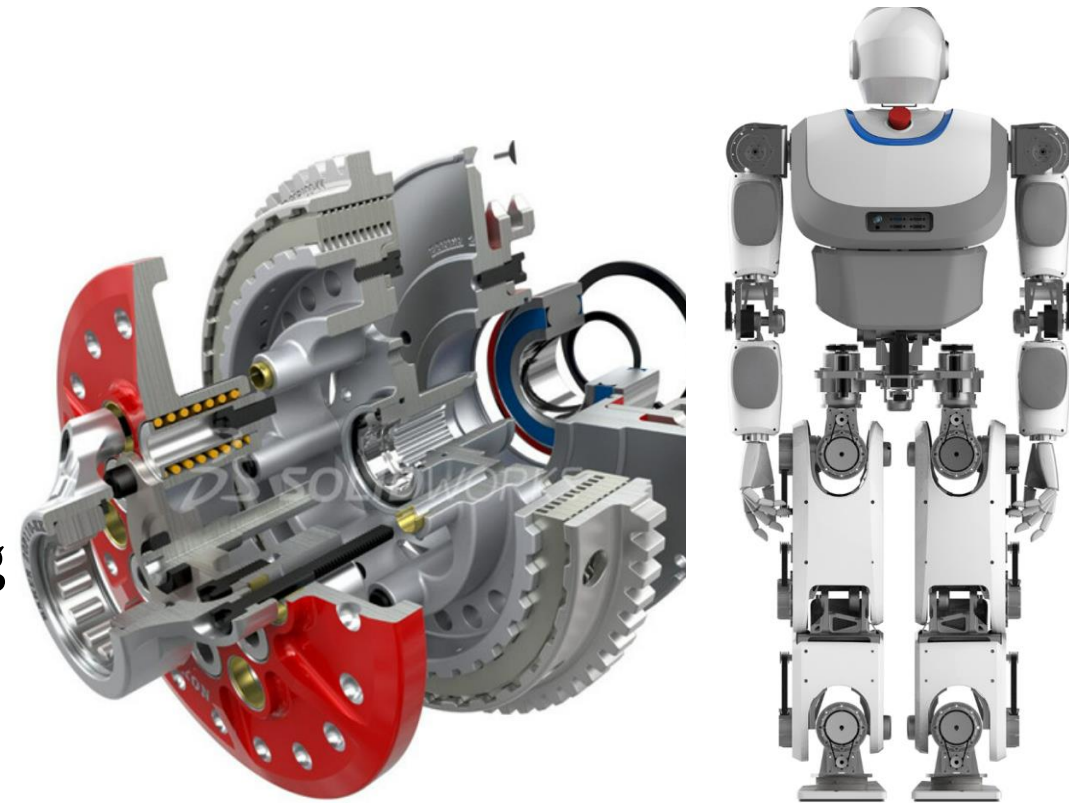
CAD (Computer Aided Design) and Engineering Drawing

2024年 秋季

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Southern University of Science and Technology



Chapter 4 Basic Solids and Intersections

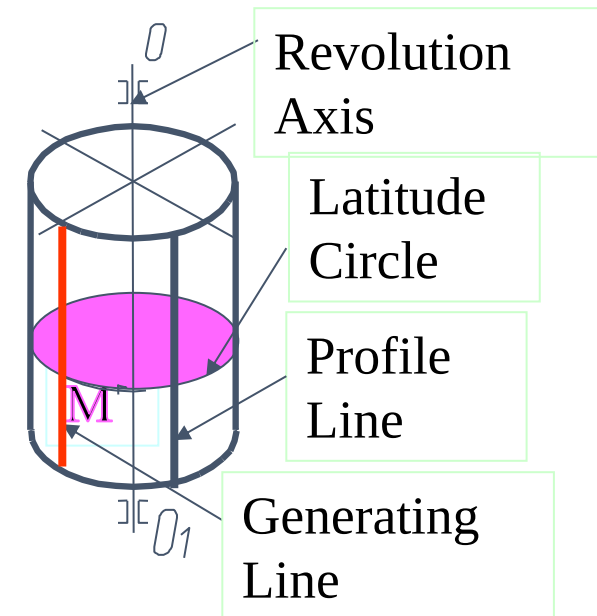
- 4-1 Basic Plane Solids
- 4-2 Intersections of Basic Plane Solids
- 4-3 Basic Solids of Revolution
- 4-4 Intersections of Solids of Revolution
- 4-5 Dimensioning of Basic Solids

The Concept of Solids of Revolution

- **Solid of Revolution:** A solid bounded by a **surface of revolution** or by surfaces of revolution and planes.
- **Surface of Revolution:** A curved surface formed by **rotating** a straight line or curve around a fixed axis.
- **Revolution Axis:** The fixed axis.
- **Generating Line:** The **moving** line that forms the surface.
- **Profile Line:** Any position of the **generating line** on the surface is called a profile line of the surface.
- **Latitude Circle:** The path traced by any point on the profile line is a circle, known as the latitude circle, which is perpendicular to the axis.

Common solids of revolution include:

Cylinder Cone Sphere Torus (Self-study)

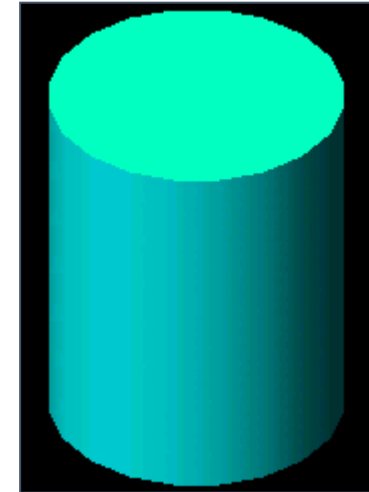
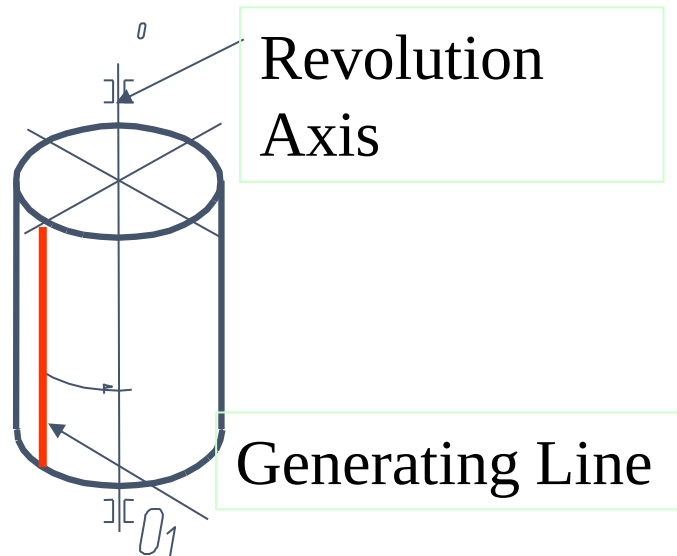


1. Cylinder

A cylinder is enclosed by **a cylindrical surface** and **two end planes**.

1) Formation of the cylindrical surface:

A straight line, when rotated around an axis for one complete revolution, forms a cylindrical surface.



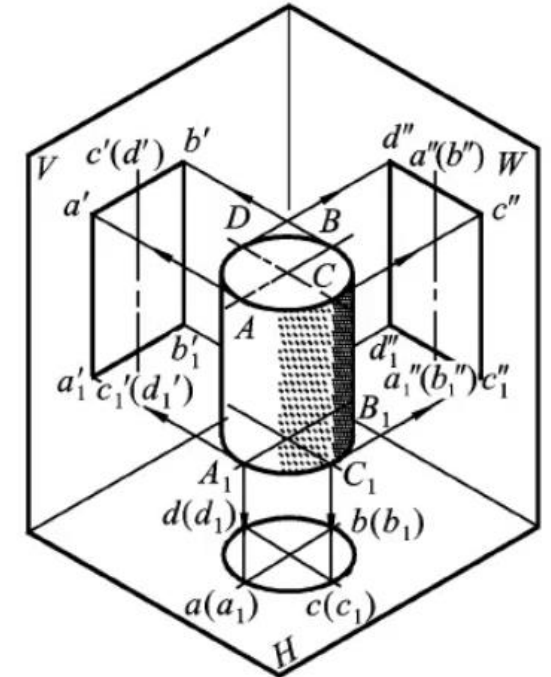
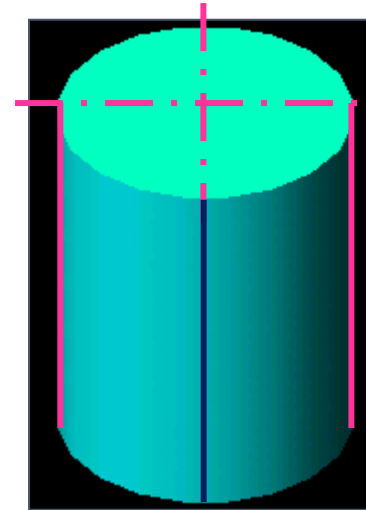
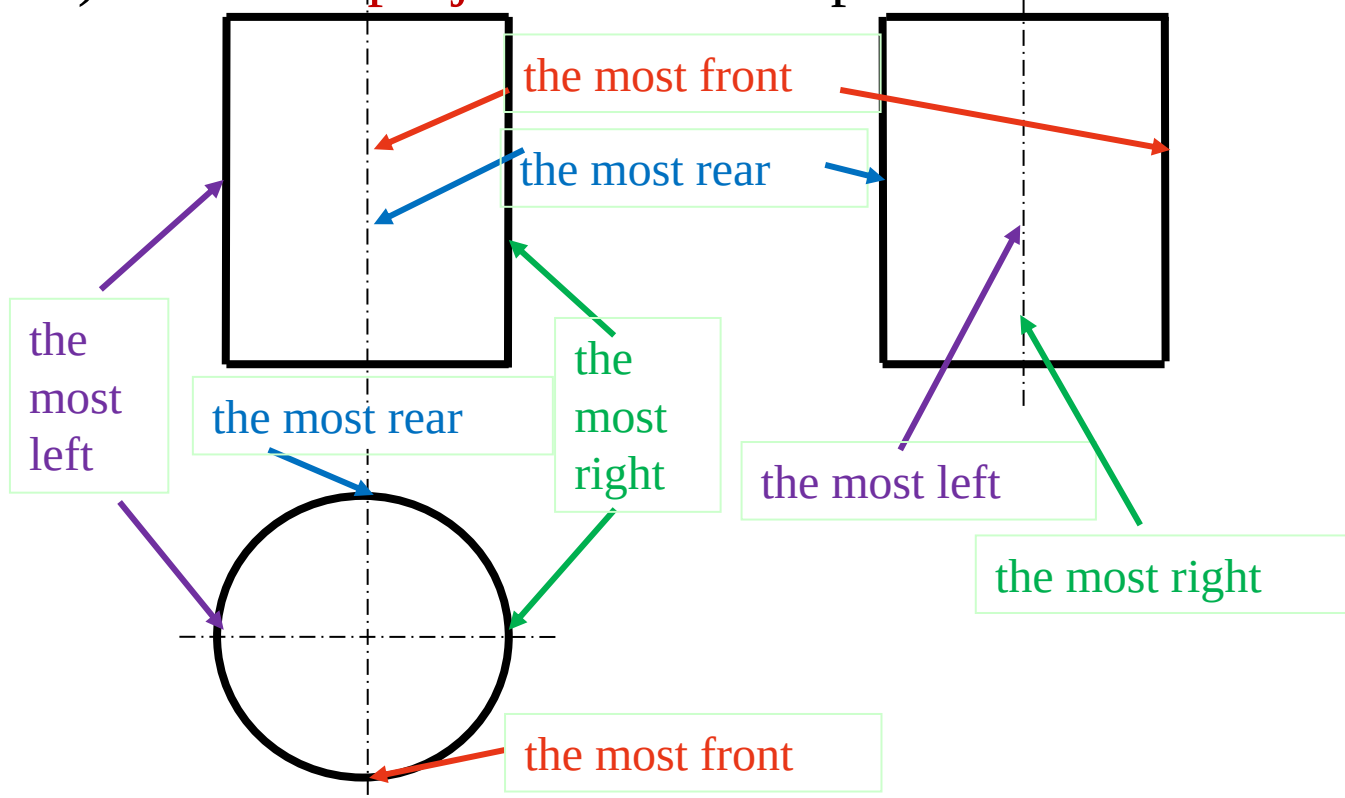
4-3 Basic Solids of Revolution



1. Cylinder

2) Three-view drawing of a cylinder

- 1) Draw the three projections of **the axis**.
- 2) Draw the three projections of **the top and bottom bases**.
- 3) Draw **the projections** of the profile lines.

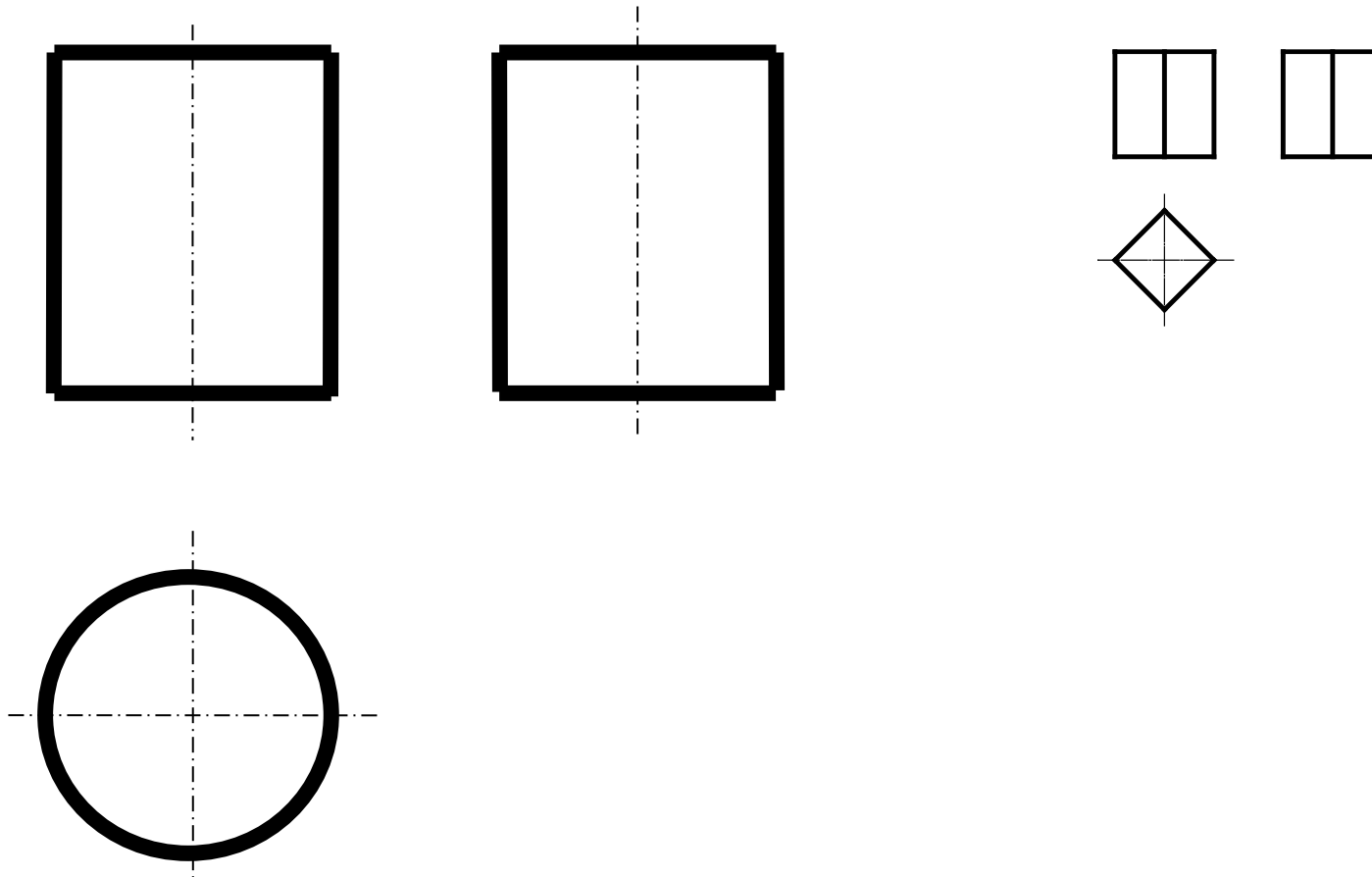


- ◆ The profile lines are **the most front, most rear, most left, and most right** profile lines of the cylindrical surface.

4-3 Basic Solids of Revolution



1. Cylinder 3) Differences between the three-view drawings of a cylinder and a pyramid



4-3 Basic Solids of Revolution

1. Cylinder

4) Visibility of the cylindrical surface

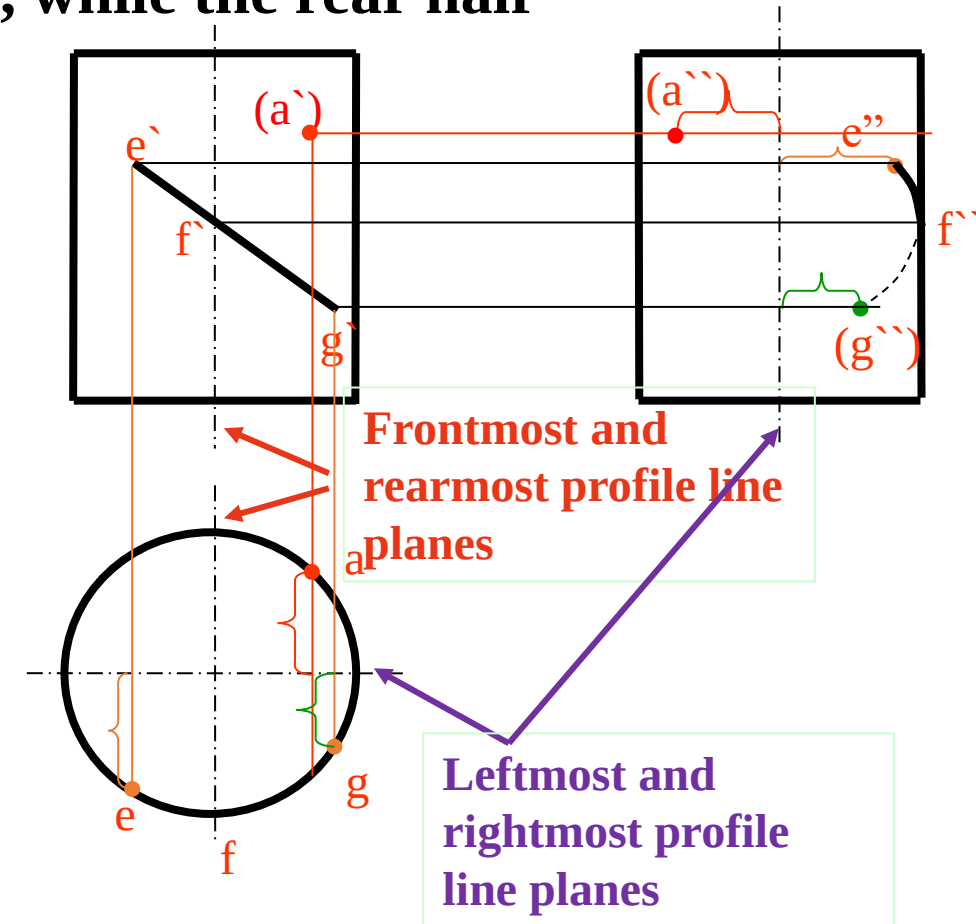
The **leftmost and rightmost profile lines** divide the cylindrical surface into **front and rear halves**. In the front view, the front half is visible, while the rear half is hidden.

The **frontmost and rearmost profile lines** divide the cylindrical surface into **left and right**. In the left view, the left half is visible, while the right half is hidden.

● Point on a cylindrical surface

Example (1): Given a' , find a and a''

(2): Given the front projection $e'g'$ of line EG on the cylindrical surface, find the other two projections.

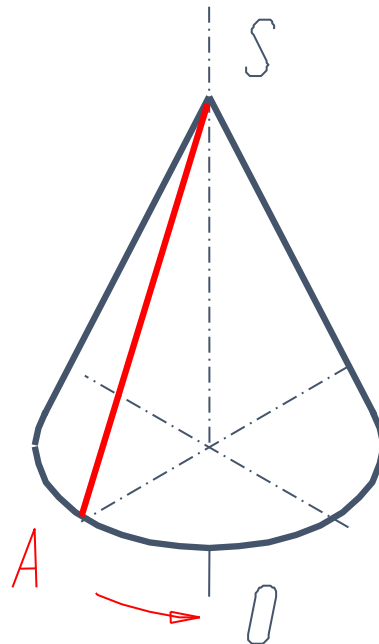


2. Cone

A **cone** is enclosed by a conical surface and a base.

1) Formation of the conical surface:

A straight line, when rotated around another straight line with which it intersects for one complete revolution, forms a conical surface.



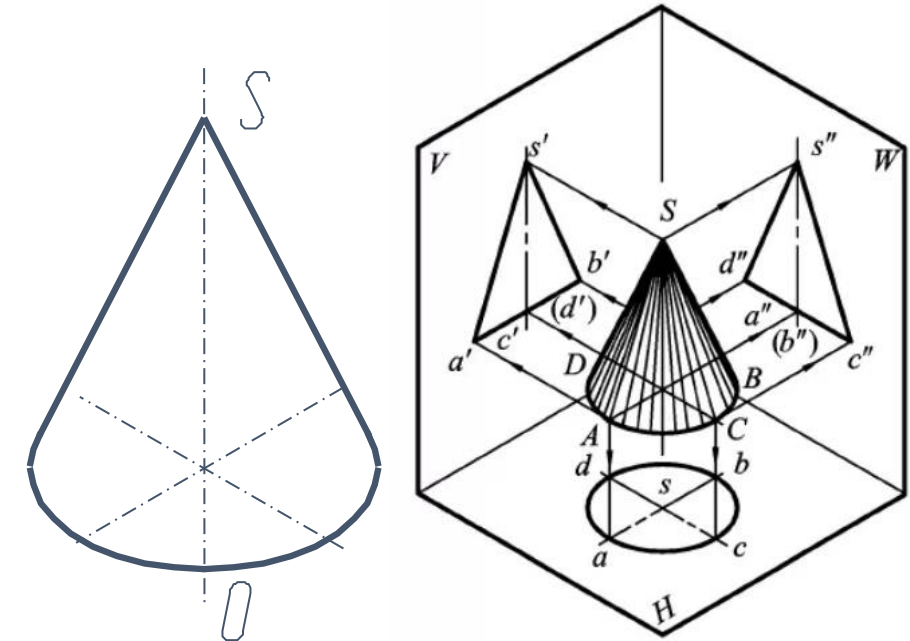
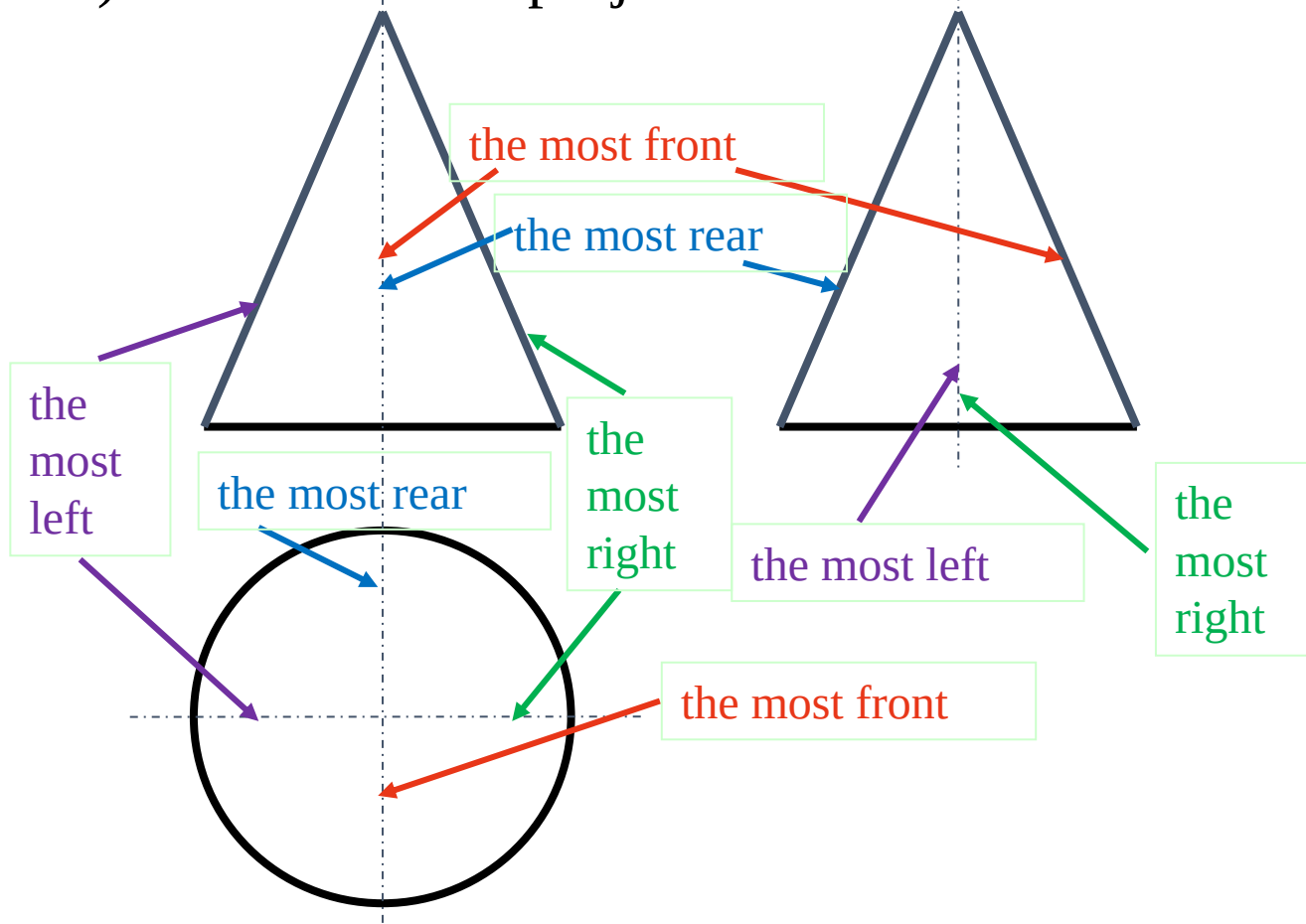
4-3 Basic Solids of Revolution



2. Cone

2) Three-view drawing of a cone

- 1) Draw the three projections of **the axis**.
- 2) Draw the three projections of **the base**.
- 3) Draw the three projections of **the conical surface**.



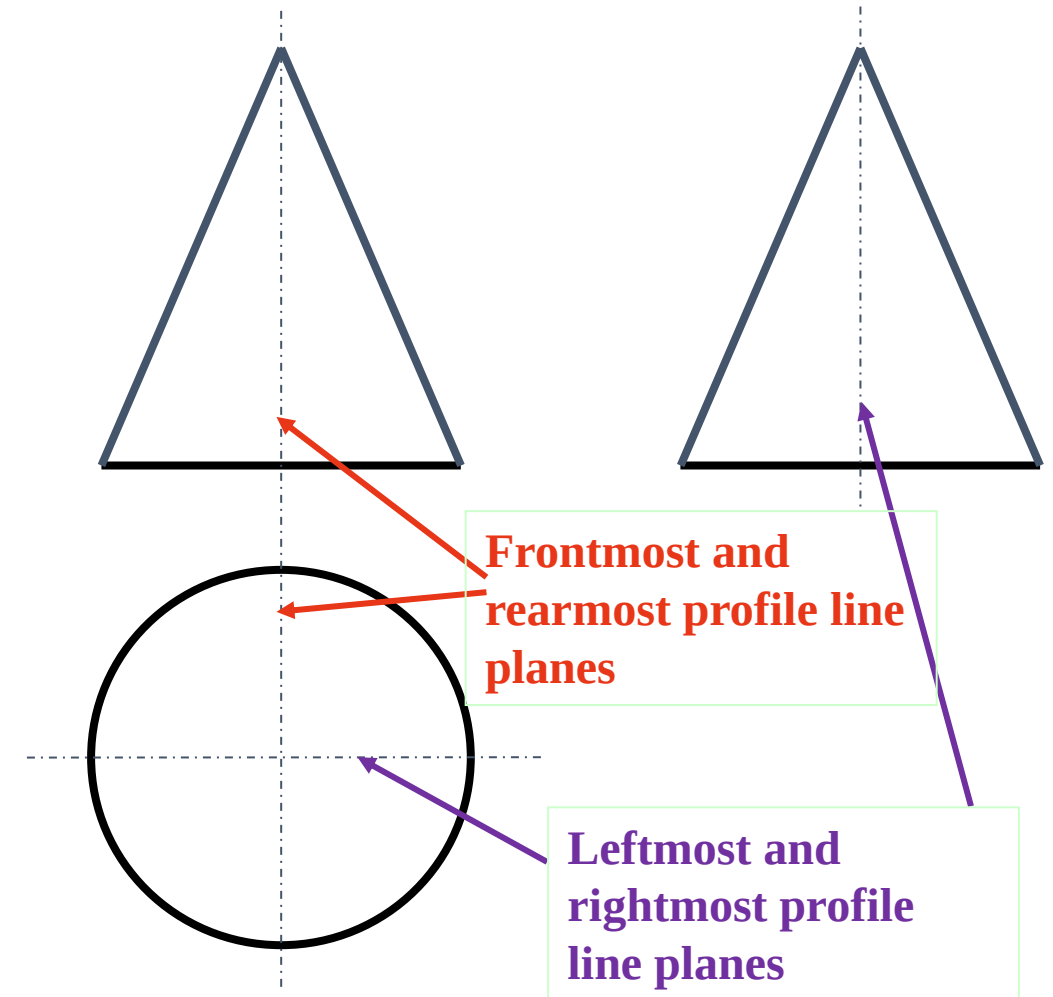
- ◆ The profile lines are **the most front, most rear, most left, and most right profile lines** of the conical surface.

2. Cone

3) Visibility of the conical surface

The planes containing **the leftmost and rightmost profile lines** divide the conical surface into **front and rear** halves. In the front view, the front half is visible, while the rear half is hidden.

The planes containing **the frontmost and rearmost profile lines** divide the conical surface into **left and right** halves. In the left view, the left half is visible, while the right half is hidden.



2. Cone

4) Point on a conical surface

Methods:

(1) Profile line method

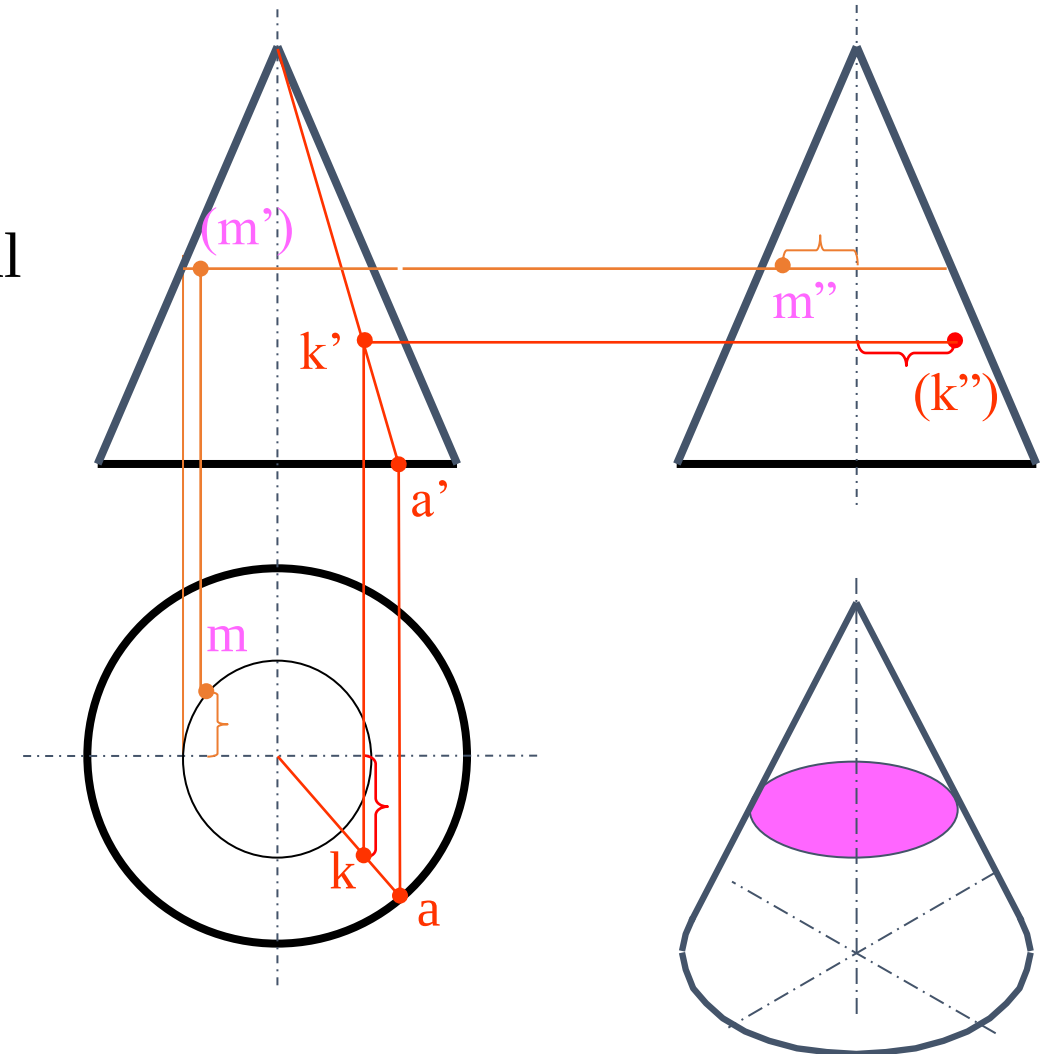
All profile lines except the four are in general position.

(2) Latitude circle method

Points equidistant from the axis (belonging to the same latitude circle) are at the same height.

Example (1): Given k' , find k and k'' .

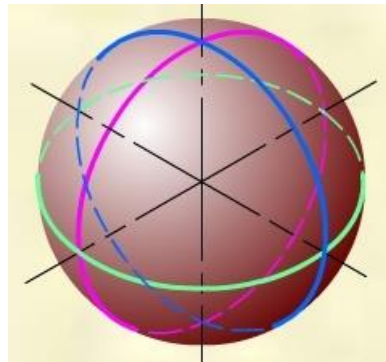
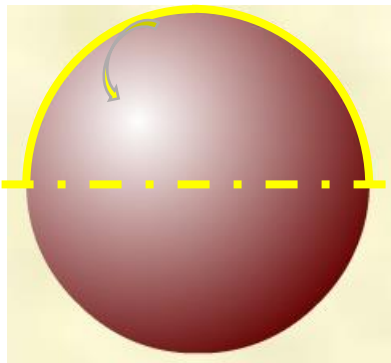
(2): Given m , find m' and m'' .



3. Sphere

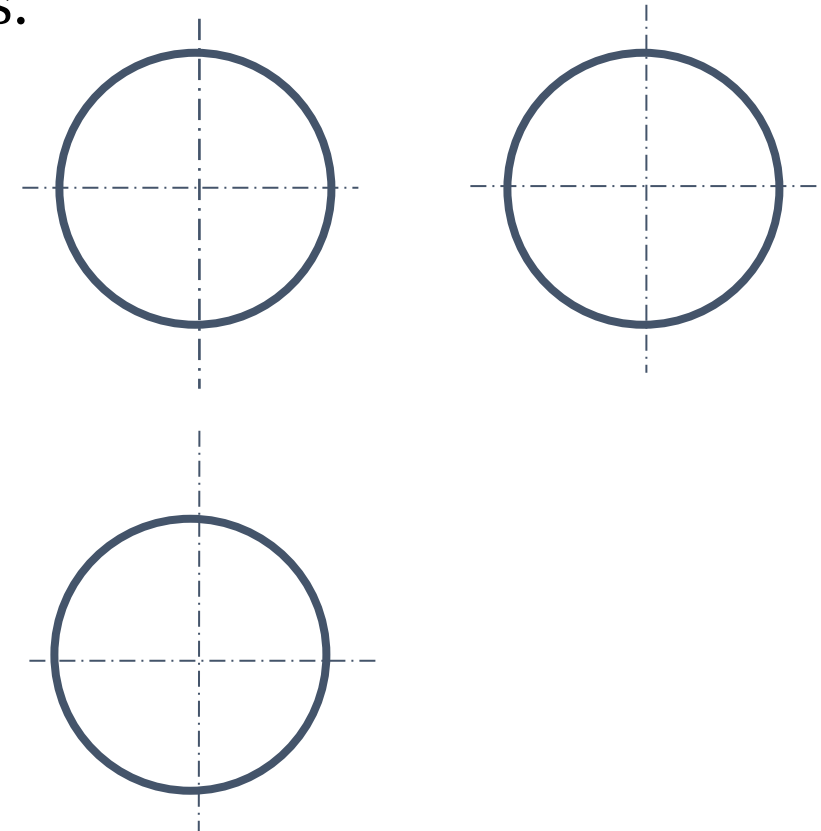
1) Formation of a sphere

A sphere can be considered as formed by rotating a semicircle around its own diameter for one complete revolution.



2) Three-view drawing of a sphere

The projections of a sphere in three directions are three circles of maximum size, with equal diameters.

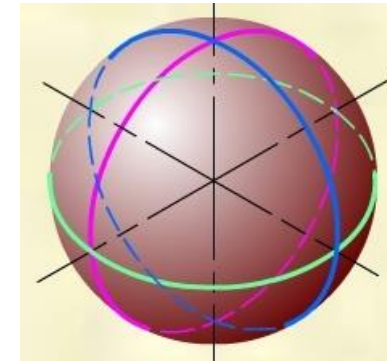
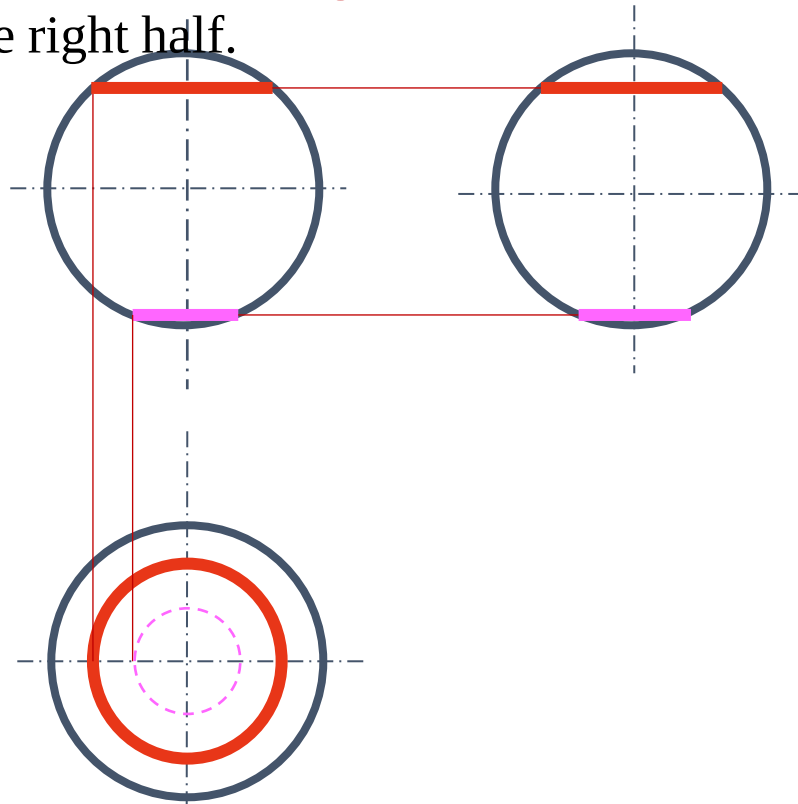


3. Sphere 3) Visibility of a sphere

In the front view, **the largest circle on the front** divides the sphere into a visible front half and an invisible rear half.

In the top view, **the largest horizontal circle** divides the sphere into a visible upper half and an invisible lower half.

In the left view, **the largest side circle** divides the sphere into a visible left half and an invisible right half.



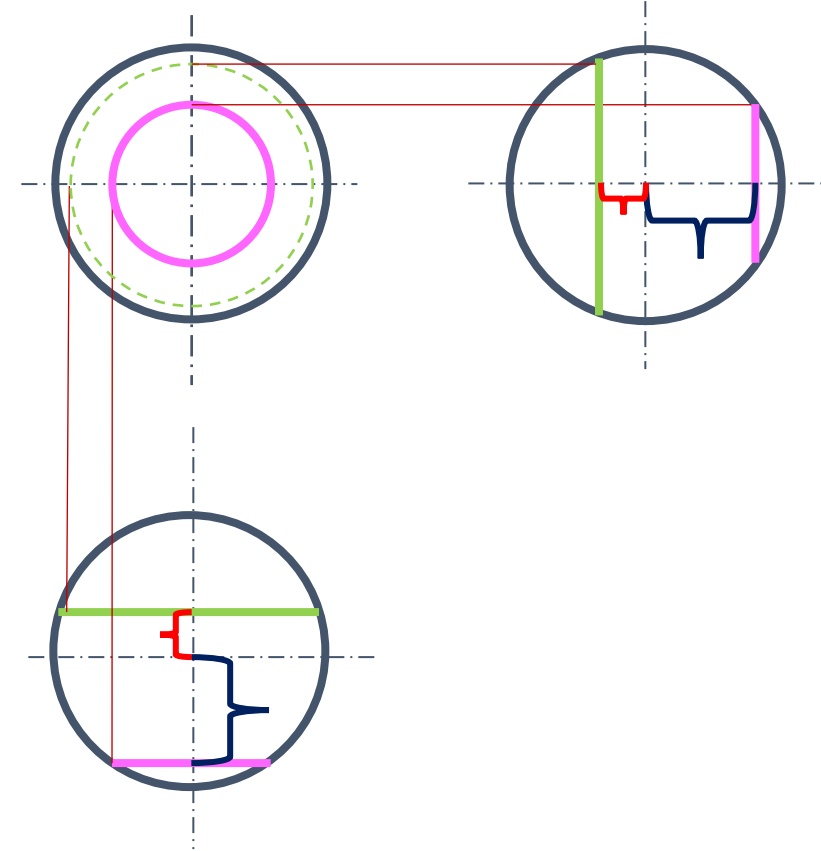
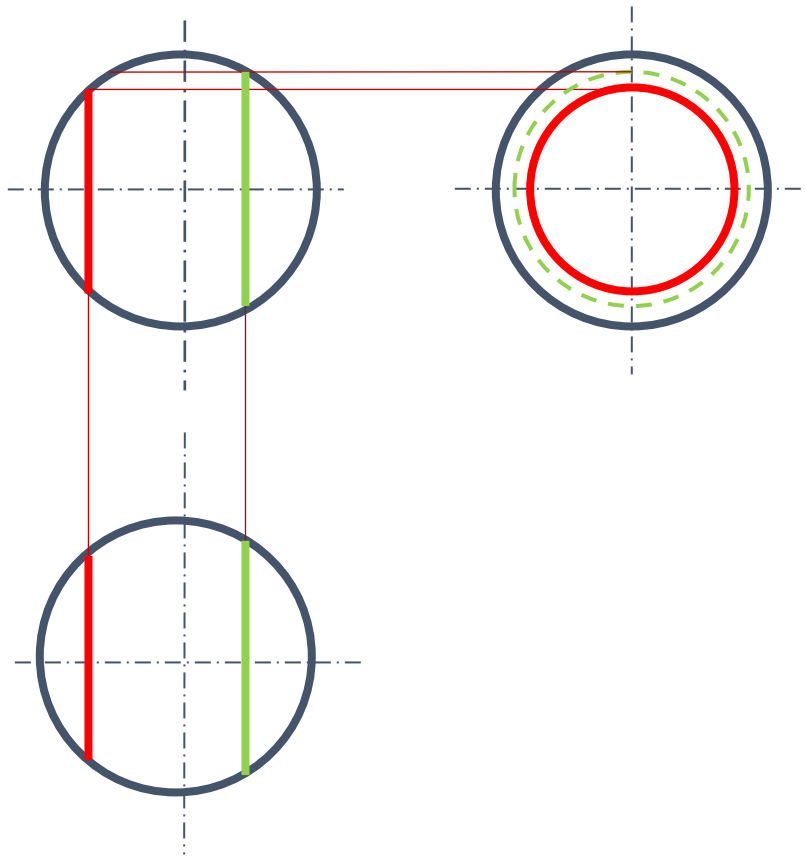
The height of the cutting plane determines the diameter of the section circle.

4-3 Basic Solids of Revolution



3. Sphere

4) Intersection of a sphere with side and front planes

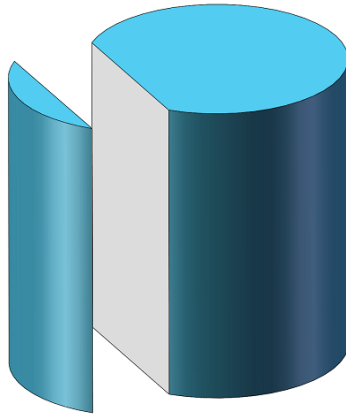


Chapter 4 Basic Solids and Intersections

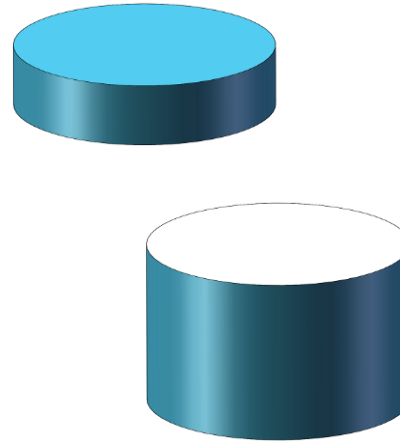
- **4-1 Basic Plane Solids**
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- **4-5 Dimensioning of Basic Solids**

1. Intersection of a Cylinder

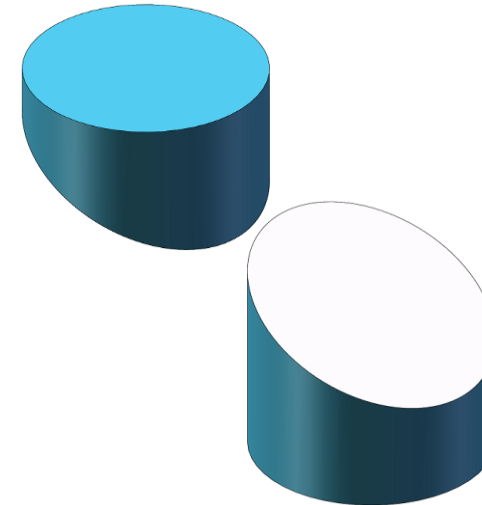
The shape of the intersection line between the cutting plane and the cylindrical surface depends on the relative position of **the cutting plane** to **the cylinder's axis**.



Parallel to the axis
(Two parallel lines)



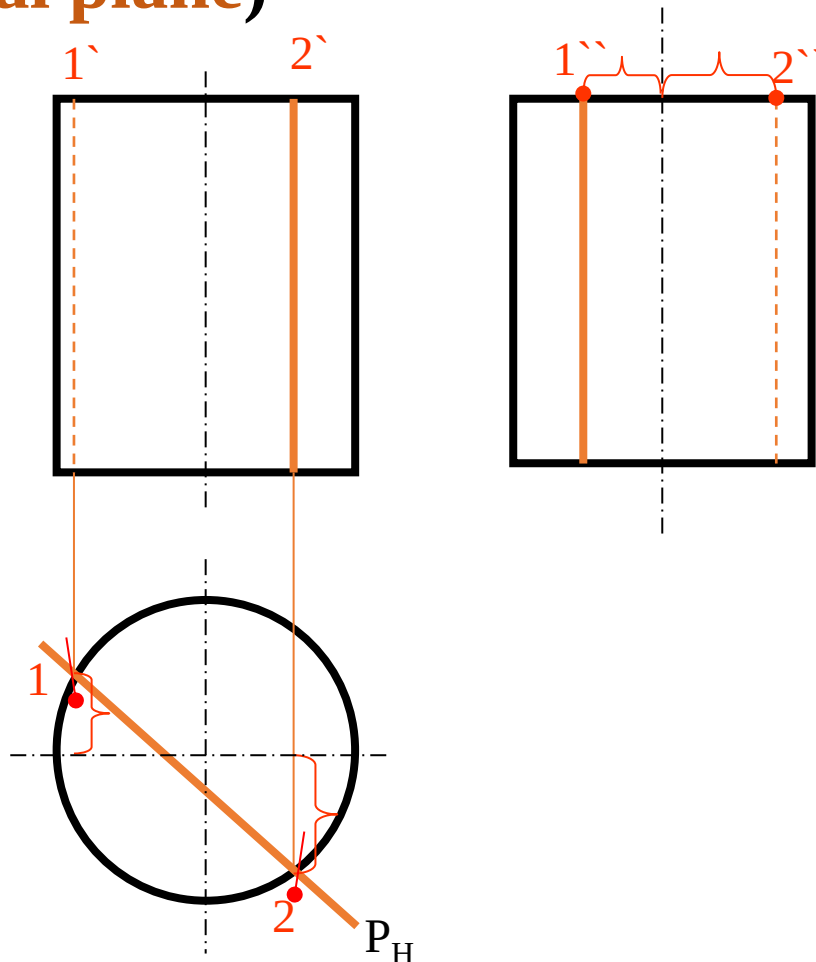
Perpendicular to
the axis
(Circle)



Oblique to the axis
(Ellipse)

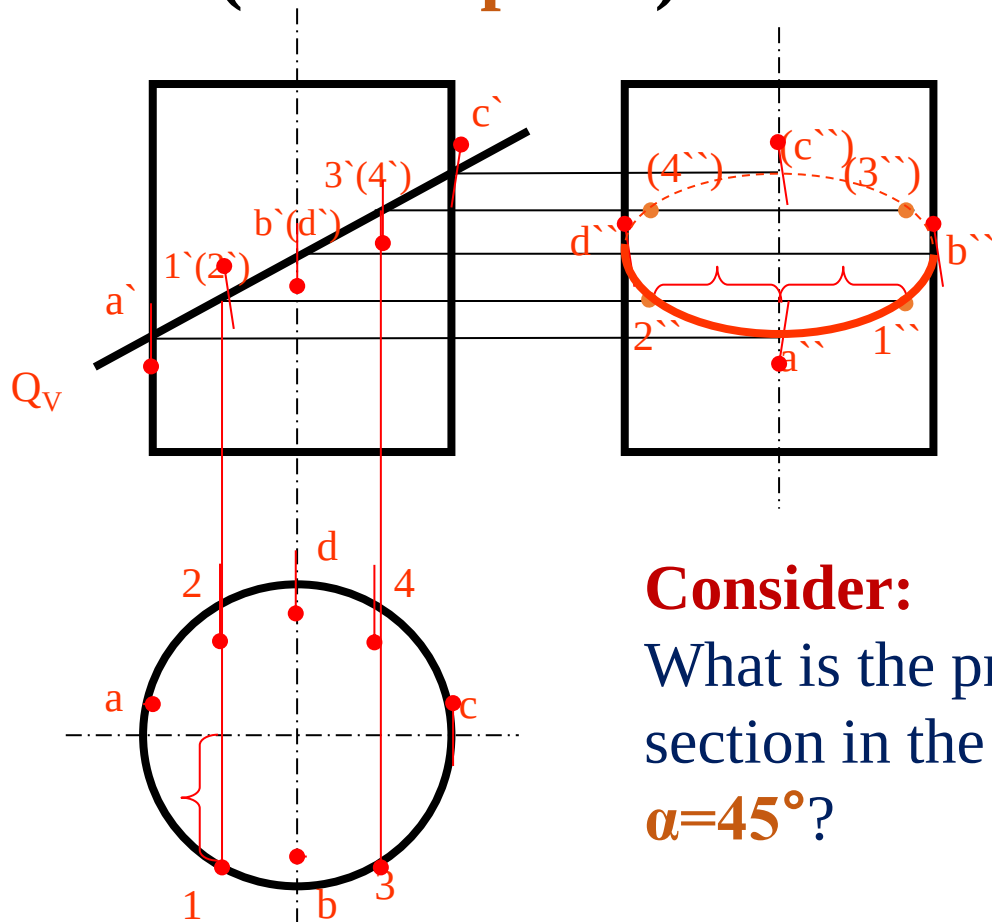
1. Intersection of a Cylinder

- 1) Cutting plane parallel to the axis (vertical plane)
2) Cutting plane perpendicular to the axis (horizontal plane, simple, skipped)



1. Intersection of a Cylinder

3) Cutting plane oblique to the axis (**frontal plane**)

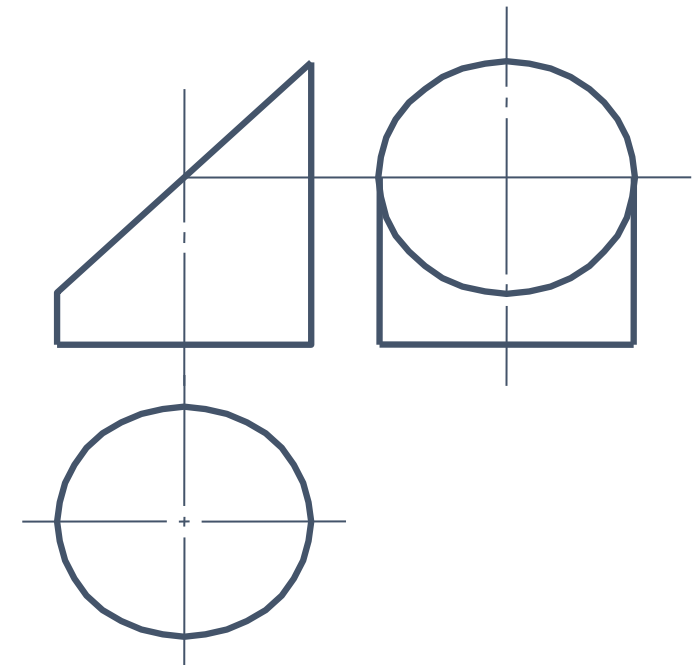


Consider:

What is the projection of the section in the left view when $\alpha=45^\circ$?

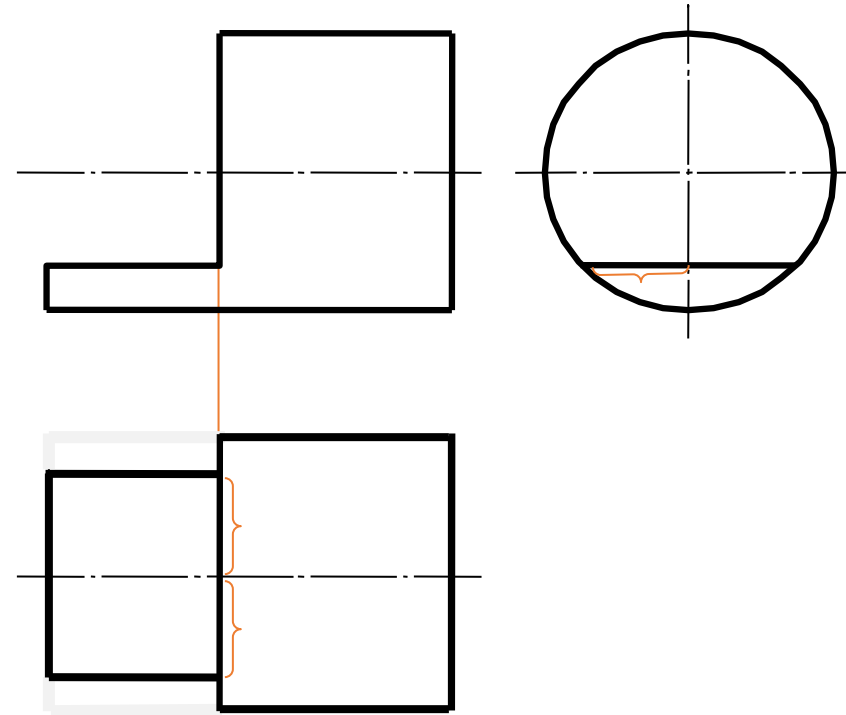
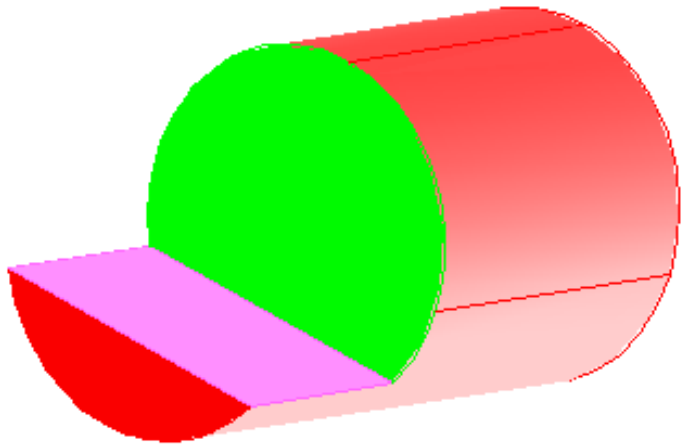
Steps for drawing:

- **Locate special points:** Intersections of Q_v with extreme profile lines; boundary points between visible and invisible parts.
- Determine **general points**.
- **Assess visibility** and connect points in sequence.



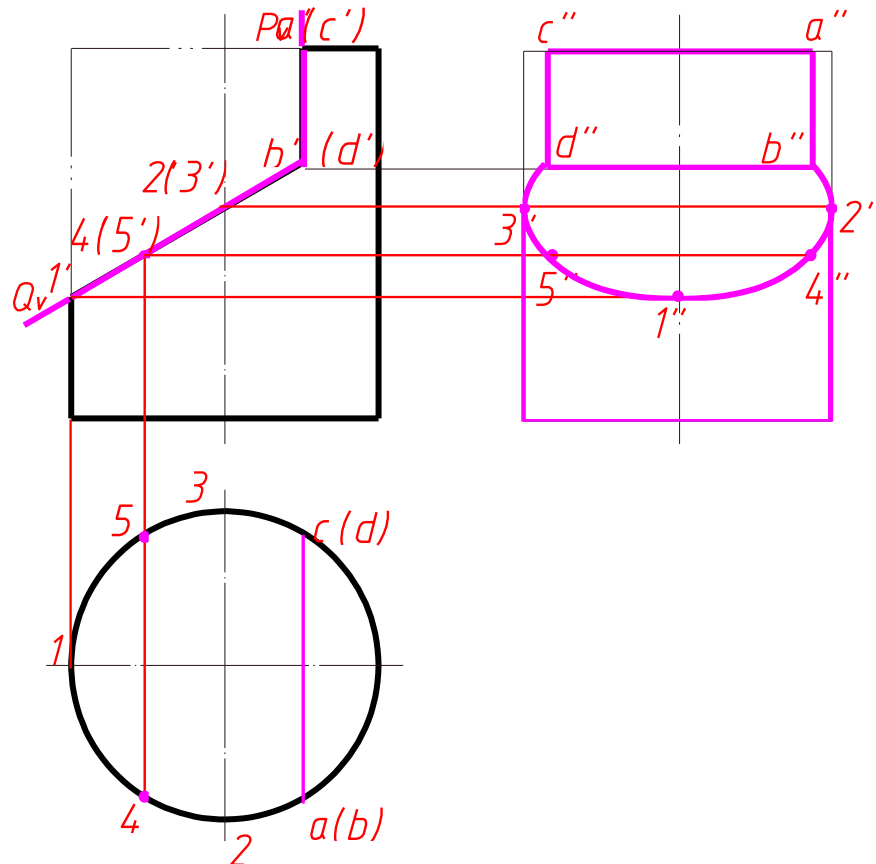
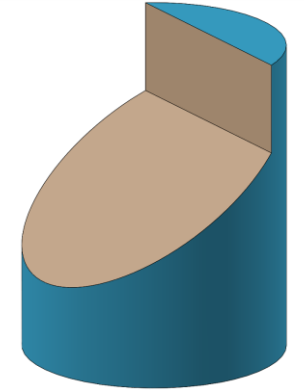
1. Intersection of a Cylinder

Example: Given the front and left views of a cylinder, find its top view.



1. Intersection of a Cylinder

Example: Complete the horizontal and side projections of the cylinder after it has been cut.



Using the accumulative property to pick points on the surface

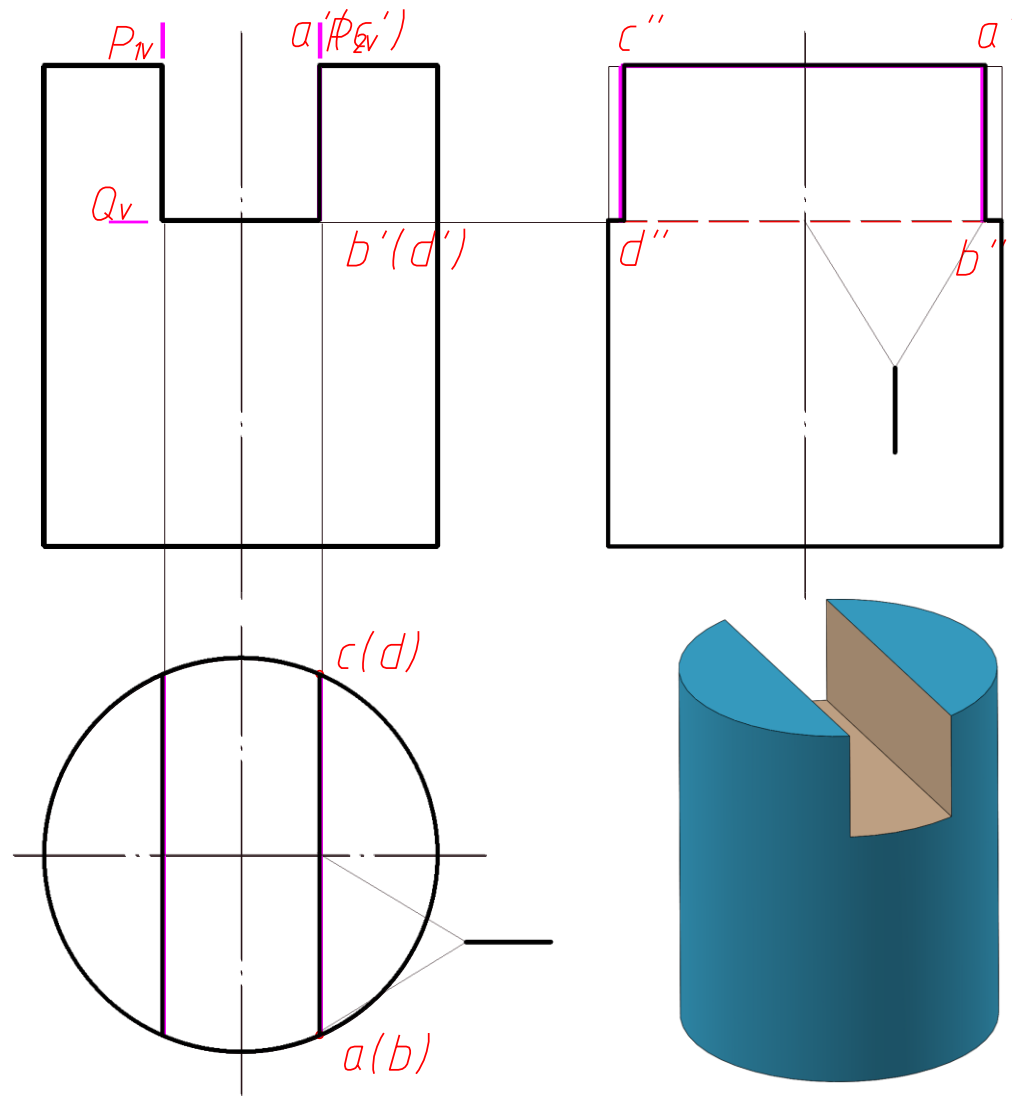
(I) Shape Analysis

- (1) Spatial shape of the intersection line
- (2) Projection of the intersection line

(II) Drawing Steps

- (1) Draw the left view of the cylinder before it is cut.
- (2) Draw the horizontal and side projections of the intersection line between the side plane and the cylinder.
- (3) Draw the side projection of the intersection line between the frontal plane and the cylinder.
- (4) Determine visibility and connect the projections of the intersection lines.
- (5) Neaten up and darken the lines.

1. Intersection of a Cylinder



Example: Given the front projection of a cylinder with a rectangular cut, find its horizontal and side projections.

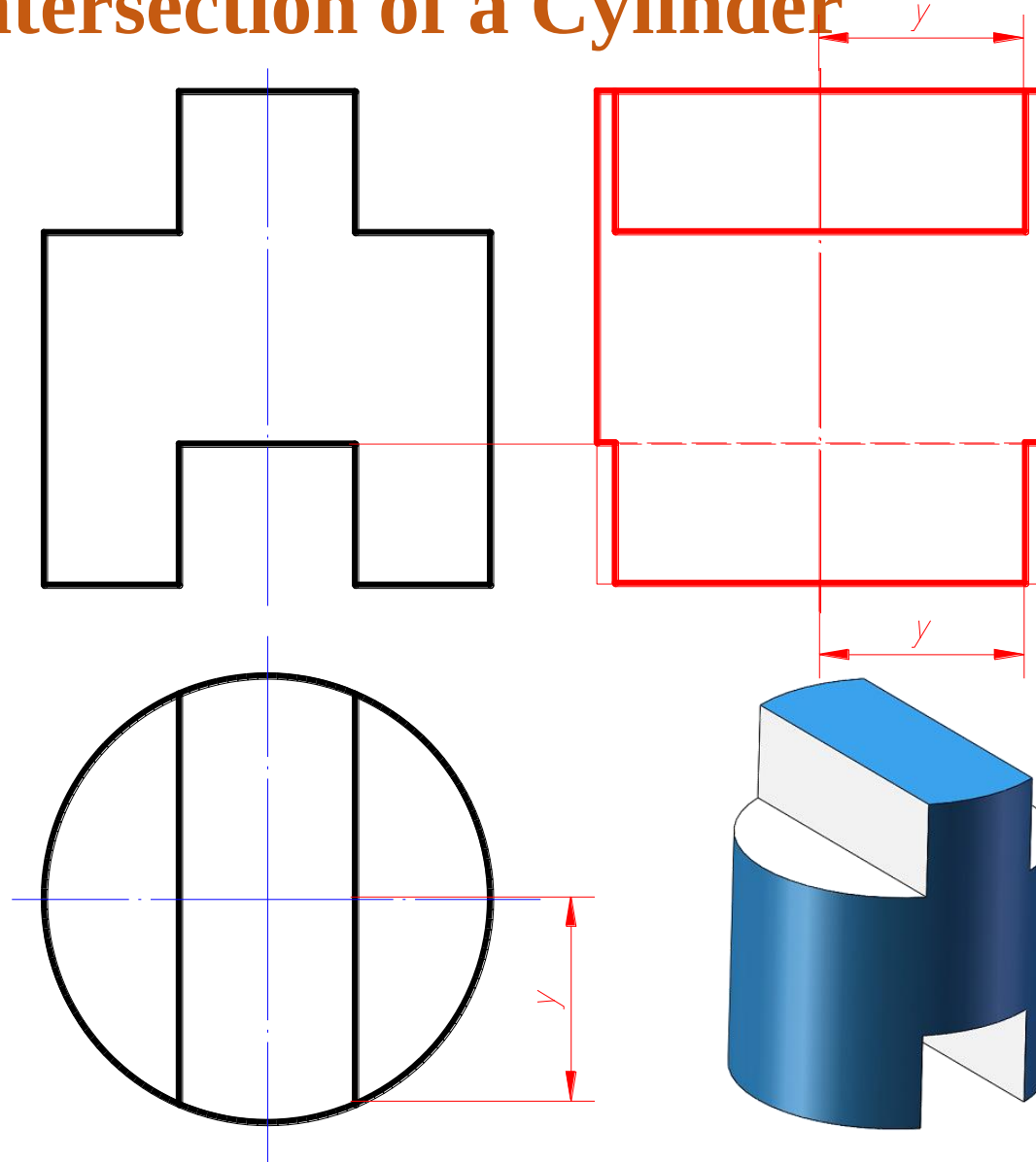
(I) Shape Analysis

- (1) Spatial shape of the intersection line
- (2) Projection of the intersection line

(II) Drawing Steps

- (1) Draw the horizontal and side projections of the cylinder before it is cut.
- (2) Find the intersection lines between the rectangular slot and the cylinder.
- (3) Erase unnecessary lines and thicken the visible contour lines.

1. Intersection of a Cylinder



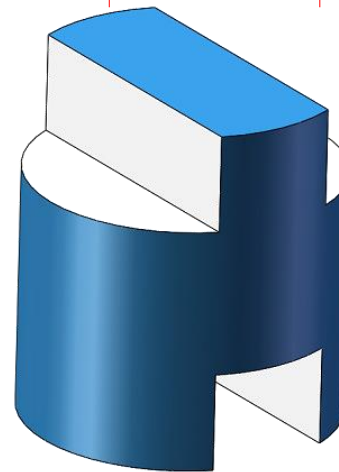
Example: Complete the left view of the cylinder after it has been cut by a plane.

(I) Shape Analysis

- (1) Spatial shape of the intersection line
- (2) Projection of the intersection line

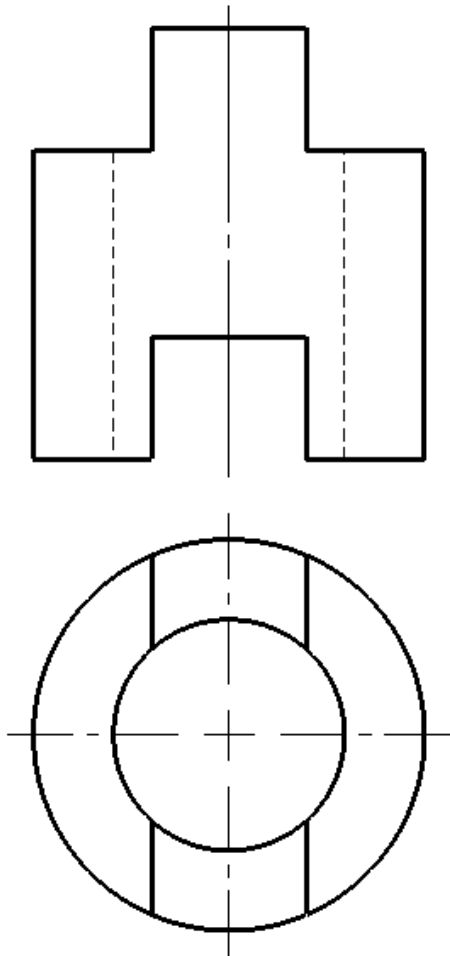
(II) Drawing Steps

- (1) Draw the left view of the cylinder before it is cut.
- (2) Find the intersection lines between the upper side plane and the horizontal plane with the cylinder.
- (3) Find the intersection lines between the lower rectangular slot and the cylinder.
- (4) Erase unnecessary lines and thicken the visible contour lines.



1. Intersection of a Cylinder

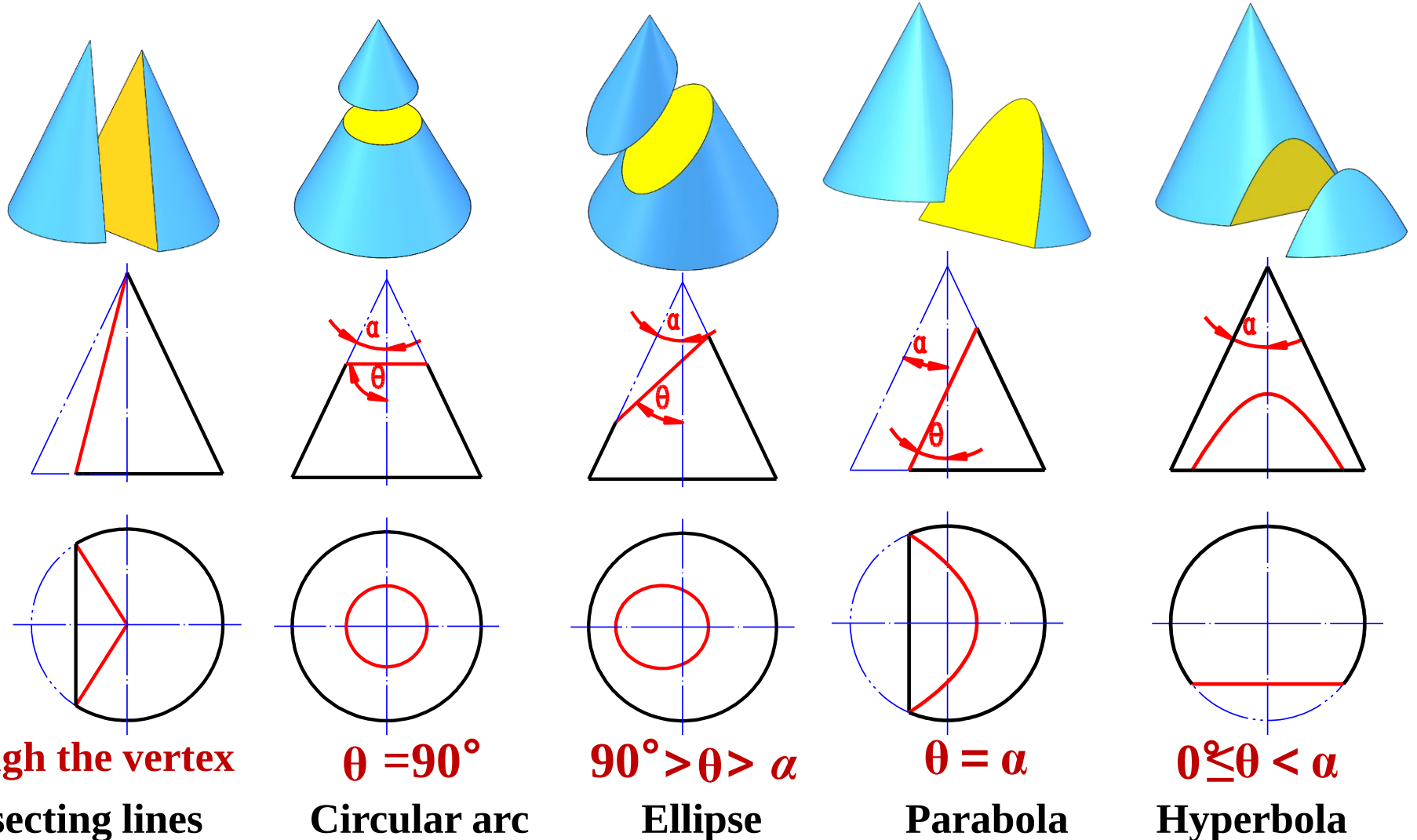
Example: The left view of a cylindrical tube after it has been cut by a plane.



Observe, analyze the differences from the previous problem, and visualize the solid.

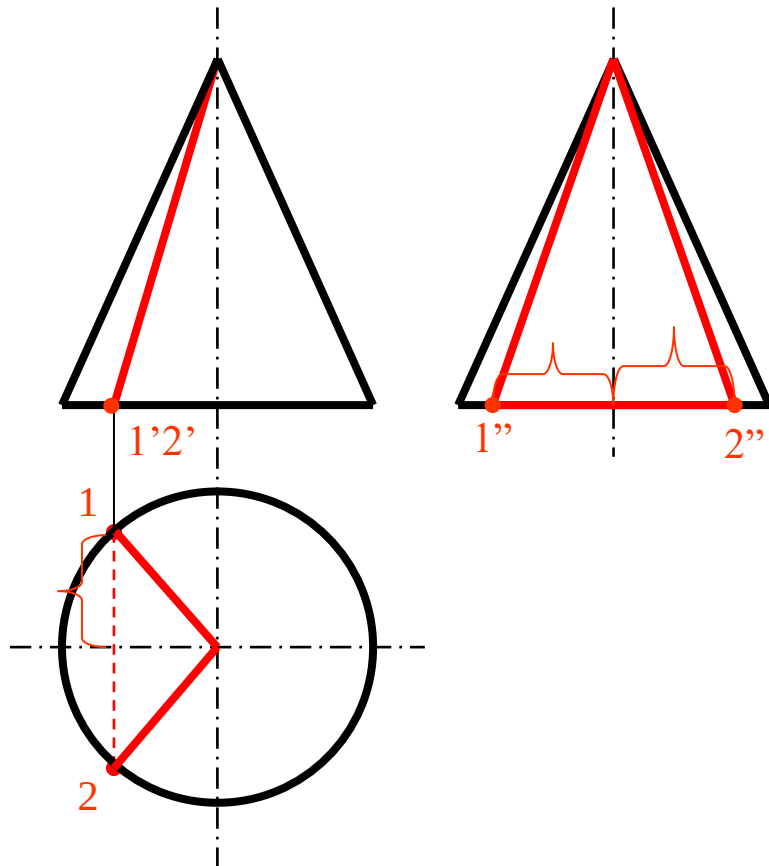
2. Intersection of a Cone

When a plane intersects a cone, depending on the position of the cutting plane relative to the cone's axis, there are five types of intersection lines.

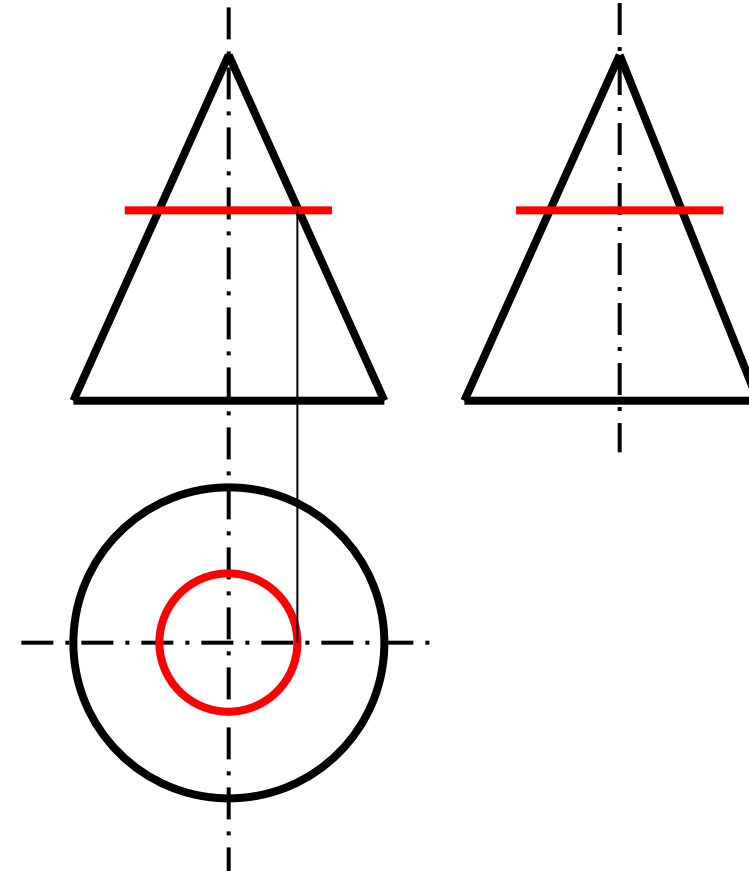


2. Intersection of a Cone

1) Cutting plane through the vertex

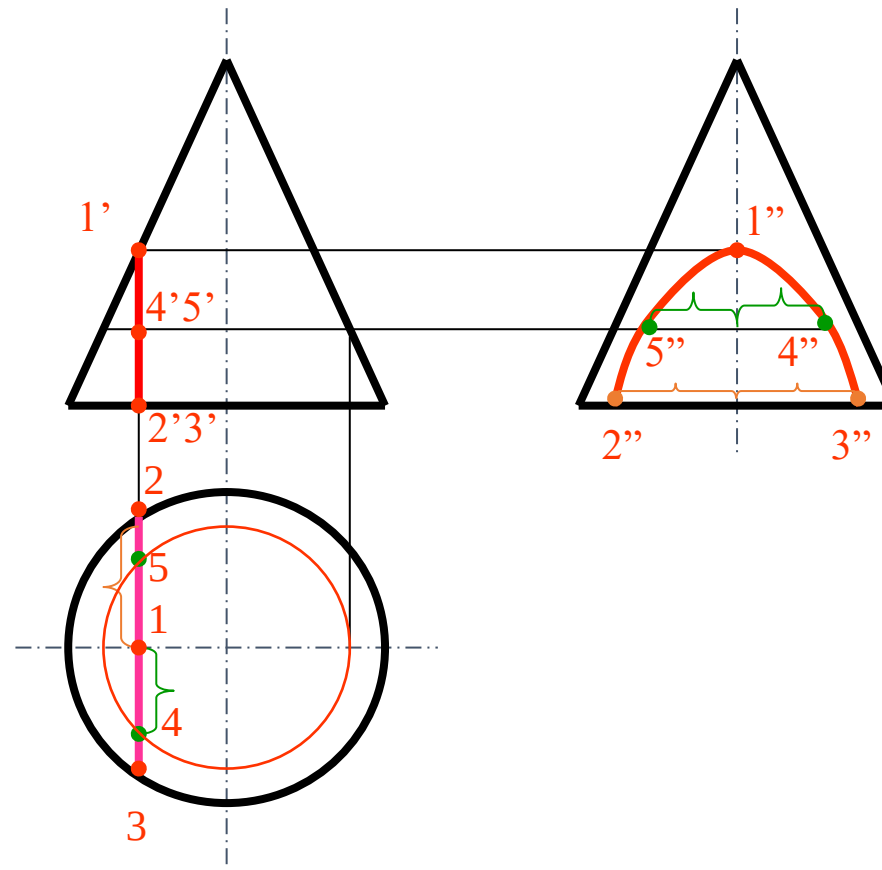


2) Cutting plane parallel to the base



2. Intersection of a Cone

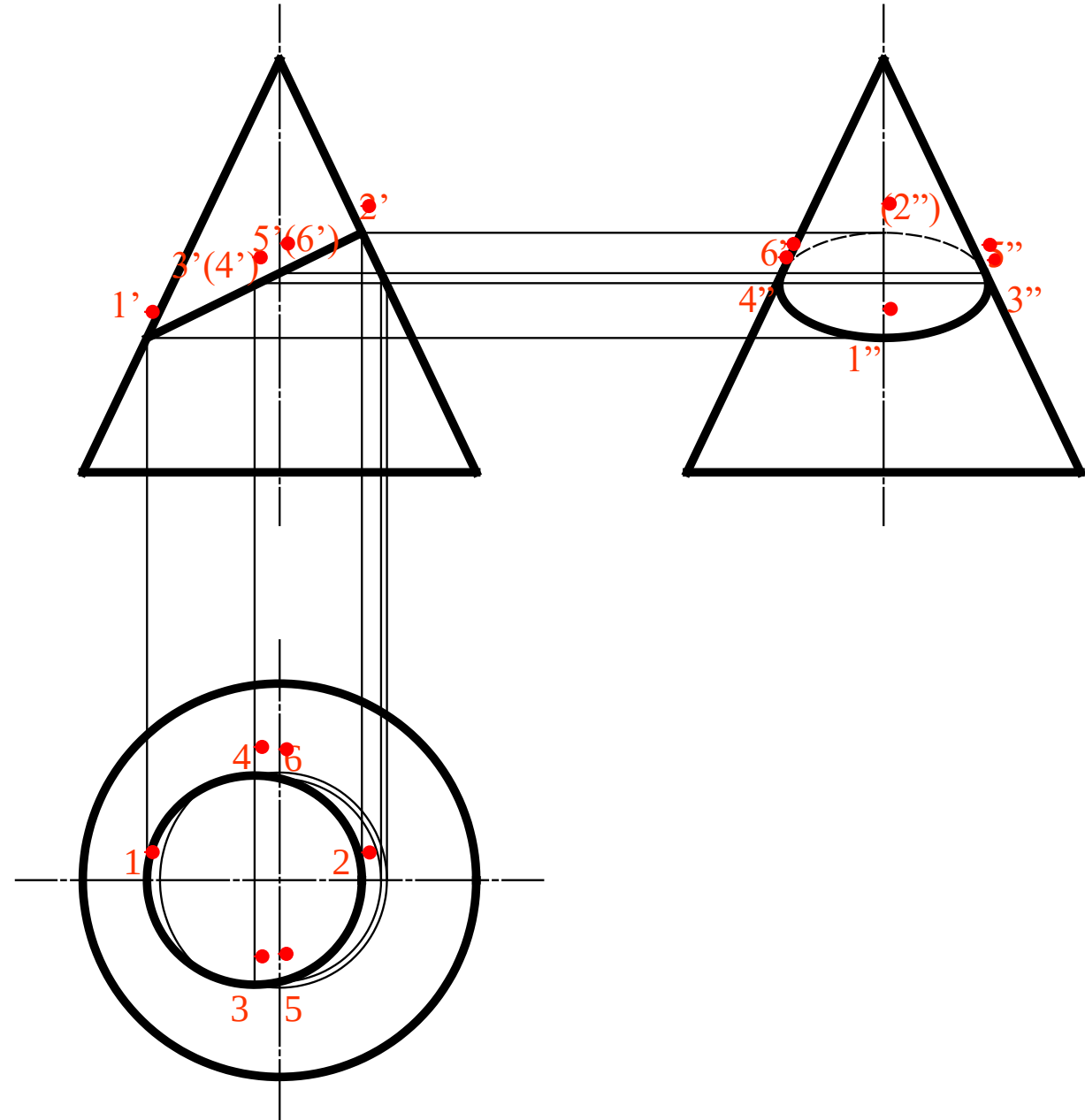
3) Cutting plane parallel to the axis (hyperbola)



2. Intersection of a Cone

4) Cutting plane oblique to the axis (ellipse)
(Just for knowledge)

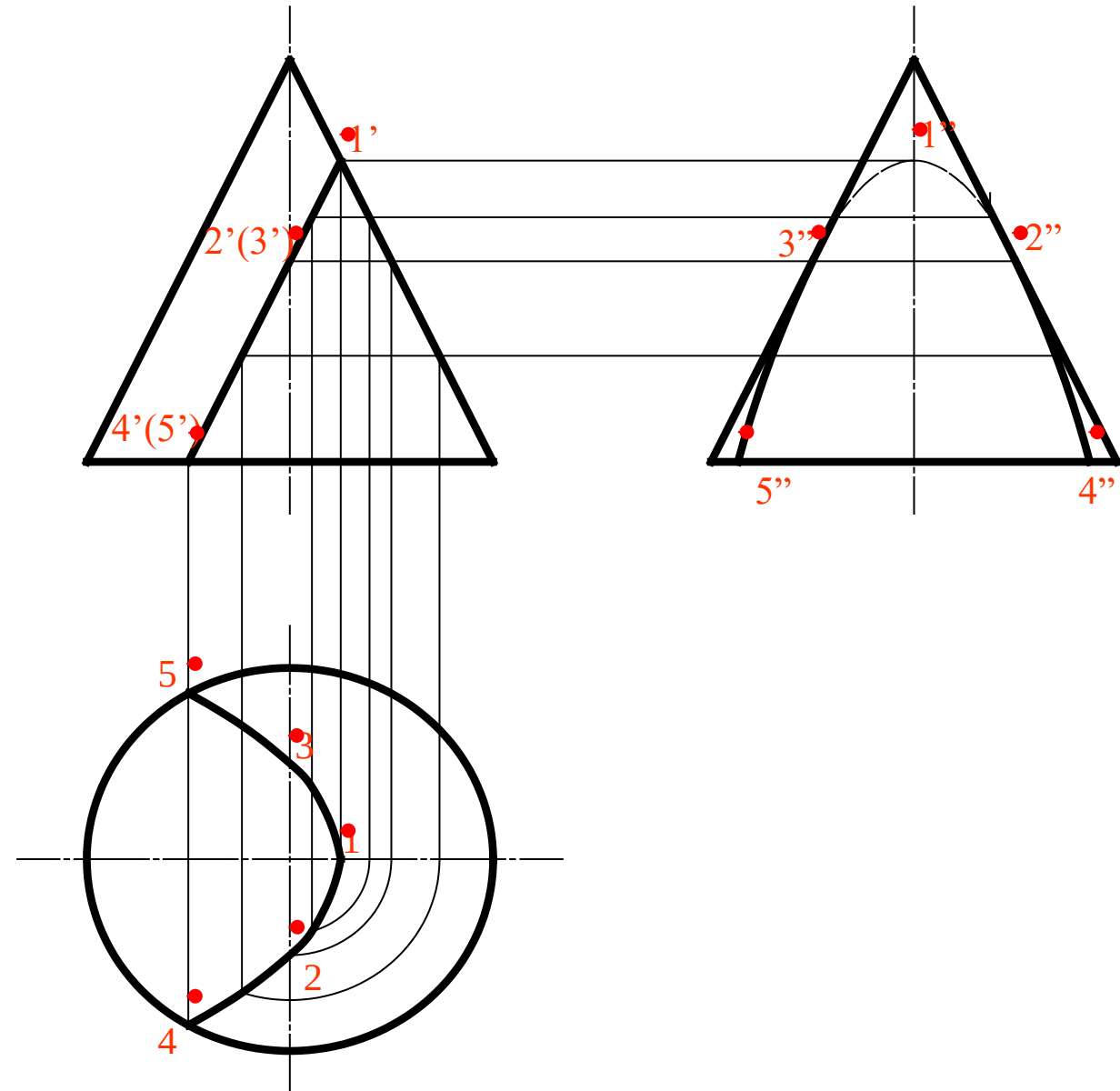
General position point: Locate the latitude circle it belongs to



2. Intersection of a Cone

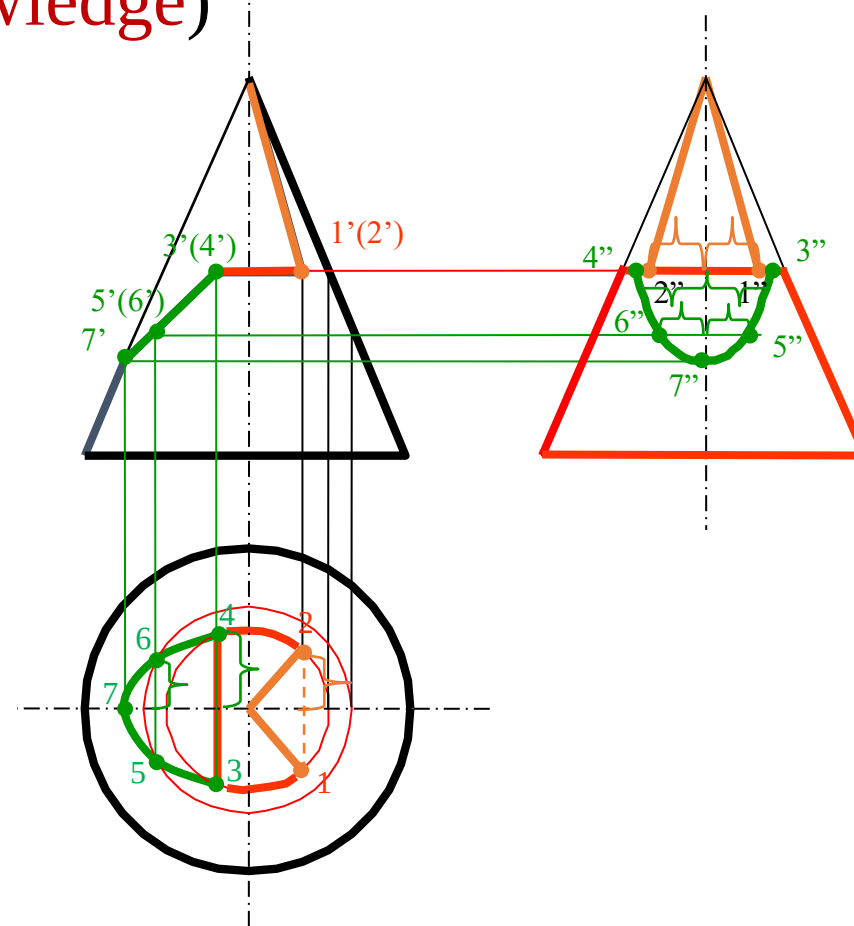
5) Cutting plane parallel
to the profile line
(parabola)
(Just for knowledge)

General position point: Locate
the latitude circle it belongs to



2. Intersection of a Cone

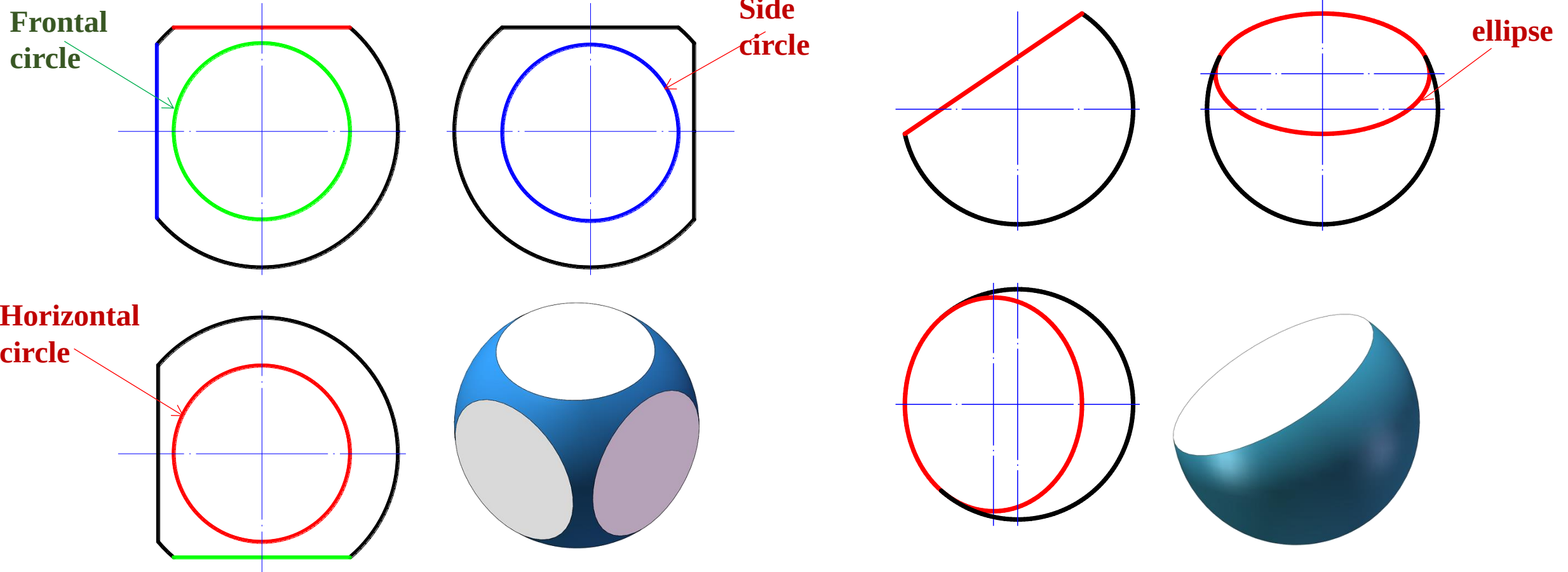
Example: Given the front view of a cone, find its top view and left view (**Just for knowledge**)



3. Intersection of a Sphere

When a plane intersects a sphere, the intersection line is always **a circle**.

Depending on the relative position of the cutting plane to the projection plane, the projection of the intersection line may be **an ellipse, a circle, or a straight line**.



Practice problems

**Let's practice together, and I'll make sure
you understand!😊**